Publications

Refereed: 6 first author (117 citations), 5 second/third author (131 citations), 61 nth author (1,945 citations)

Selected peer-reviewed journal publications (248 citations)

2026: J. K. Luhn, **R. A. Rubenzahl**, S. Halverson, & L. L. Zhao, "Exposure-averaged Gaussian Processes for Combining Overlapping Datasets", *The Astronomical Journal*, 171, 342 [1 citation]

2026: **R. A. Rubenzahl**, S. Hattori, S. Särkkä, W. M. Farr, J. K. Luhn, M. Bedell, & D. Foreman-Mackey, "Scalable Gaussian Processes for Integrated and Overlapping Measurements via Augmented State-space Models", *The Astronomical Journal*, 171, 259 [2 citations]

2025: S. Giacalone, A. W. Howard, **R. A. Rubenzahl**, F. Dai, L. B. Handley, H. Isaacson, S. Halverson, M. Brodheim, M. Brown, T. W. Carmichael, et al., "A Hot Jupiter with a Retrograde Orbit Around A Sun-like Star and a Toy Model of Hot Jupiters in Wide Binary Star Systems", *Publications of the Astronomical Society of the Pacific*, 137, 074401 [5 citations]

2025: L. B. Handley, A. W. Howard, **R. A. Rubenzahl**, F. Dai, D. Tyler, R. A. Lee, S. Giacalone, H. Isaacson, B. Fulton, A. Householder, et al., "An Obliquity Measurement of the Hot Neptune TOI-1694b", *The Astronomical Journal*, 169, 212 [14 citations]

2024: **R. A. Rubenzahl**, F. Dai, S. Halverson, A. W. Howard, A. Householder, B. Fulton, A. Behmard, S. R. Gibson, A. Roy, A. P. Shaum, et al., "KPF Confirms a Polar Orbit for KELT-18 b", *The Astronomical Journal*, 168, 188 [10 citations]

2024: **R. A. Rubenzahl**, A. W. Howard, S. Halverson, C. Petrovich, I. Angelo, G. Stefánsson, F. Dai, A. Householder, B. Fulton, S. R. Gibson, et al., "Obliquity Constraints for the Extremely Eccentric Sub-Saturn Kepler-1656 b", *The Astrophysical Journal Letters*, 971, L40 [8 citations]

2024: **R. A. Rubenzahl**, F. Dai, A. W. Howard, J. J. Lissauer, J. Van Zandt, C. Beard, S. Giacalone, J. M. A. Murphy, A. Chontos, J. Lubin, et al., "The TESS-Keck Survey. XII. A Dense 1.8 $R_\oplus$ Ultra-short-period Planet Possibly Clinging to a High-mean-molecular-weight Atmosphere after the First Gigayear", *The Astronomical Journal*, 167, 153 [12 citations]

2023: **R. A. Rubenzahl**, S. Halverson, J. Walawender, G. M. Hill, A. W. Howard, M. Brown, E. Ida, J. Tehero, B. J. Fulton, S. R. Gibson, et al., "Staring at the Sun with the Keck Planet Finder: An Autonomous Solar Calibrator for High Signal-to-noise Sun-as-a-star Spectra", *Publications of the Astronomical Society of the Pacific*, 135, 125002 [30 citations]

2022: J. C. Siegel, **R. A. Rubenzahl**, S. Halverson, & A. W. Howard, "Into the Depths: A New Activity Metric for High-precision Radial Velocity Measurements Based on Line Depth Variations", *The Astronomical Journal*, 163, 260 [13 citations]

2021: **R. A. Rubenzahl**, F. Dai, A. W. Howard, A. Chontos, S. Giacalone, J. Lubin, L. J. Rosenthal, H. Isaacson, N. M. Batalha, I. J. M. Crossfield, et al., "The TESS-Keck Survey. IV. A Retrograde, Polar Orbit for the Ultra-low-density, Hot Super-Neptune WASP-107b", *The Astronomical Journal*, 161, 119 [55 citations]

2021: C. Piaulet, B. Benneke, **R. A. Rubenzahl**, A. W. Howard, E. J. Lee, D. Thorngren, R. Angus, M. Peterson, J. E. Schlieder, M. Werner, et al., "WASP-107b's Density Is Even Lower: A Case Study for the Physics of Planetary Gas Envelope Accretion and Orbital Migration", *The Astronomical Journal*, 161, 70 [98 citations]

Peer-reviewed publications contributed to as collaborator (191 citations)

2026: K. L. Hobbs, C. A. Watson, J. C. Costes, Y. Unruh, D. C. Yaptangco, et al. **(incl. RAR)**, "The Fe I 4377 Å line as a solar faculae indicator: insights from spectral ratio analysis", *Monthly Notices of the Royal Astronomical Society*, 551, stag1462

2026: S. Blunt, J. Wang, R. Murray-Clay, B. Macintosh, **R. A. Rubenzahl**, et al., "Evidence for a Peak at ∼0.3 in the Eccentricity Distribution of Typical Super-Jovian Exoplanets", *The Astrophysical Journal Letters*, 999, L26 [6 citations]

2025: A. Behmard, C. L. Brinkman, S. Hattori, **R. A. Rubenzahl**, & M. Bedell, "A Link between Rocky Planet Density and Host Star Chemistry", *The Astronomical Journal*, 170, 282 [8 citations]

2025: E. Y. Zhang, H.-Y. Teng, F. Dai, A. W. Howard, S. P. Halverson, et al. **(incl. RAR)**, "TOI-880 is an Aligned, Coplanar, Multiplanet System", *The Astronomical Journal*, 170, 175 [5 citations]

2025: A. W. Howard, E. Sinukoff, S. Blunt, E. A. Petigura, I. J. M. Crossfield, et al. **(incl. RAR)**, "Planet Masses, Radii, and Orbits from NASA's K2 Mission", *The Astrophysical Journal Supplement Series*, 278, 52 [24 citations]

2024: M. Hon, D. Huber, Y. Li, T. S. Metcalfe, T. R. Bedding, et al. **(incl. RAR)**, "Asteroseismology of the Nearby K Dwarf σ Draconis Using the Keck Planet Finder and TESS", *The Astrophysical Journal*, 975, 147 [22 citations]

2024: H.-Y. Teng, F. Dai, A. W. Howard, H. Isaacson, **R. A. Rubenzahl**, et al., "The ∼50 Myr Old TOI-942c is Likely on an Aligned, Coplanar Orbit and Losing Mass", *The Astronomical Journal*, 168, 194 [10 citations]

2024: S. Giacalone, F. Dai, J. J. Zanazzi, A. W. Howard, C. D. Dressing, et al. **(incl. RAR)**, "The OATMEAL Survey. I. Low Stellar Obliquity in the Transiting Brown Dwarf System GPX-1", *The Astronomical Journal*, 168, 189 [13 citations]

2024: B. J. Hord, E. M.-R. Kempton, T. M. Evans-Soma, D. W. Latham, D. R. Ciardi, et al. **(incl. RAR)**, "Identification of the Top TESS Objects of Interest for Atmospheric Characterization of Transiting Exoplanets with JWST", *The Astronomical Journal*, 167, 233 [41 citations]

2023: S. Blunt, A. Carvalho, T. J. David, C. Beichman, J. K. Zink, et al. **(incl. RAR)**, "Overfitting Affects the Reliability of Radial Velocity Mass Estimates of the V1298 Tau Planets", *The Astronomical Journal*, 166, 62 [57 citations]

2020: A. Albert, R. Alfaro, C. Alvarez, J. C. Arteaga-Velázquez, K. P. Arunbabu, et al. **(incl. RAR)**, "Constraints on the Emission of Gamma-Rays from M31 with HAWC", *The Astrophysical Journal*, 893, 16 [5 citations]

TESS-Keck Survey peer-reviewed publications (557 citations)

I contributed substantially to the TKS infrastructure by co-developing the sample construction algorithm (Chontos et al., 2022, AJ), constructing the nightly target lists (298 calendar nights), and co-leading the RV fitting of the HIRES/APF data for all TKS systems (Polanski et al., 2024, ApJS).

2025: J. Van Zandt, et al. **(incl. RAR)**, "The TESS-Keck Survey. XXIV. Outer Giants May Be More Prevalent in the Presence of Inner Small Planets", *The Astronomical Journal*, 169, 235 [17 citations]

2024: D. Pidhorodetska, et al. **(incl. RAR)**, "The TESS-Keck Survey. XXII. A Sub-Neptune Orbiting TOI-1437", *The Astronomical Journal*, 168, 135 [3 citations]

2024: S. Lange, et al. **(incl. RAR)**, "The TESS-Keck Survey. VII. A Superdense Sub-Neptune Orbiting TOI-1824", *The Astronomical Journal*, 167, 282 [4 citations]

2024: A. S. Polanski, et al. **(incl. RAR)**, "The TESS-Keck Survey. XX. 15 New TESS Planets and a Uniform RV Analysis of All Survey Targets", *The Astrophysical Journal Supplement Series*, 272, 32 [72 citations]

2024: A. Desai, et al. **(incl. RAR)**, "The TESS-Keck Survey. XVIII. A Sub-Neptune and Spurious Long-period Signal in the TOI-1751 System", *The Astronomical Journal*, 167, 194 [5 citations]

2024: M. L. Hill, et al. **(incl. RAR)**, "The TESS-Keck Survey. XIX. A Warm Transiting Sub-Saturn-mass Planet and a Nontransiting Saturn-mass Planet Orbiting a Solar Analog", *The Astronomical Journal*, 167, 151 [3 citations]

2024: C. Beard, et al. **(incl. RAR)**, "The TESS-Keck Survey. XVII. Precise Mass Measurements in a Young, High-multiplicity Transiting Planet System Using Radial Velocities and Transit Timing Variations", *The Astronomical Journal*, 167, 70 [17 citations]

2024: A. Chontos, et al. **(incl. RAR)**, "The TESS-Keck Survey XXI: 13 New Planets and Homogeneous Properties for 21 Subgiant Systems", arXiv:2402.07893 [15 citations]

2023: J. M. Akana Murphy, et al. **(incl. RAR)**, "The TESS-Keck Survey. XVI. Mass Measurements for 12 Planets in Eight Systems", *The Astronomical Journal*, 166, 153 [18 citations]

2023: M. G. MacDougall, et al. **(incl. RAR)**, "The TESS-Keck Survey. XV. Precise Properties of 108 TESS Planets and Their Host Stars", *The Astronomical Journal*, 166, 33 [30 citations]

2023: J. Van Zandt, et al. **(incl. RAR)**, "TESS-Keck Survey. XIV. Two Giant Exoplanets from the Distant Giants Survey", *The Astronomical Journal*, 165, 60 [19 citations]

2022: M. G. MacDougall, et al. **(incl. RAR)**, "The TESS-Keck Survey. XIII. An Eccentric Hot Neptune with a Similar-mass Outer Companion around TOI-1272", *The Astronomical Journal*, 164, 97 [8 citations]

2022: E. V. Turtelboom, et al. **(incl. RAR)**, "The TESS-Keck Survey. XI. Mass Measurements for Four Transiting Sub-Neptunes Orbiting K Dwarf TOI-1246", *The Astronomical Journal*, 163, 293 [17 citations]

2022: A. Chontos, et al. **(incl. RAR)**, "The TESS-Keck Survey: Science Goals and Target Selection", *The Astronomical Journal*, 163, 297 [51 citations]

2022: P. A. Dalba, et al. **(incl. RAR)**, "The TESS-Keck Survey. VIII. Confirmation of a Transiting Giant Planet on an Eccentric 261 Day Orbit with the Automated Planet Finder Telescope", *The Astronomical Journal*, 163, 61 [38 citations]

2022: J. Lubin, et al. **(incl. RAR)**, "TESS-Keck Survey. IX. Masses of Three Sub-Neptunes Orbiting HD 191939 and the Discovery of a Warm Jovian plus a Distant Substellar Companion", *The Astronomical Journal*, 163, 101 [38 citations]

2021: M. G. MacDougall, et al. **(incl. RAR)**, "The TESS-Keck Survey. VI. Two Eccentric Sub-Neptunes Orbiting HIP-97166", *The Astronomical Journal*, 162, 265 [16 citations]

2021: N. Scarsdale, et al. **(incl. RAR)**, "TESS-Keck Survey. V. Twin Sub-Neptunes Transiting the Nearby G Star HD 63935", *The Astronomical Journal*, 162, 215 [16 citations]

2021: F. Dai, et al. **(incl. RAR)**, "TKS X: Confirmation of TOI-1444b and a Comparative Analysis of the Ultra-short-period Planets with Hot Neptunes", *The Astronomical Journal*, 162, 62 [44 citations]

2021: L. M. Weiss, et al. **(incl. RAR)**, "The TESS-Keck Survey. II. An Ultra-short-period Rocky Planet and Its Siblings Transiting the Galactic Thick-disk Star TOI-561", *The Astronomical Journal*, 161, 56 [50 citations]

2020: F. Dai, et al. **(incl. RAR)**, "The TESS-Keck Survey. III. A Stellar Obliquity Measurement of TOI-1726 c", *The Astronomical Journal*, 160, 193 [33 citations]

2020: P. A. Dalba, et al. **(incl. RAR)**, "The TESS-Keck Survey. I. A Warm Sub-Saturn-mass Planet and a Caution about Stray Light in TESS Cameras", *The Astronomical Journal*, 159, 241 [43 citations]

Peer-reviewed publications contributed to as observer (1,197 citations)

I have contributed to a number of works as a result of observing on Keck/HIRES, Keck/KPF, and through my membership on the KPF Instrument Team.

2026: E. Y. Zhang, F. Dai, A. W. Howard, S. P. Halverson, H. Isaacson, et al. **(incl. RAR)**, "Spin-Orbit Alignment of Two Neptune-size Planets Younger Than 500 Myr: TOI-560 b and TOI-5082 b", *The Astronomical Journal*, 172, 201

2026: A. Householder, F. Dai, A. Kesseli, A. W. Howard, S. Halverson, et al. **(incl. RAR)**, "The Keck Planet Finder SURFS-UP Survey. I. Transmission Spectroscopy of WASP-76 b", *The Astronomical Journal*, 172, 48 [5 citations]

2026: M.-T. Wang, F. Dai, H.-G. Liu, K. Masuda, A. W. Howard, et al. **(incl. RAR)**, "TOI-4495: A Pair of Aligned, Near-resonant Sub-Neptunes That Likely Experienced Overstable Migration", *The Astronomical Journal*, 171, 135 [5 citations]

2025: C. L. Brinkman, L. M. Weiss, D. Huber, R. A. Lee, J. Kolecki, et al. **(incl. RAR)**, "The Compositions of Rocky Planets in Close-in Orbits Tend to Be Earth-like", *The Astronomical Journal*, 170, 109 [22 citations]

2025: H.-Y. Teng, F. Dai, A. W. Howard, S. Halverson, H. Isaacson, et al. **(incl. RAR)**, "Stellar Obliquity of the Ultra-short-period Planet System HD 93963", *The Astronomical Journal*, 170, 51 [7 citations]

2025: R. A. Lee, F. Dai, A. W. Howard, S. Halverson, J. Gomez Barrientos, et al. **(incl. RAR)**, "TOI-6324 b: An Earth-mass Ultra-short-period Planet Transiting a Nearby M Dwarf", *The Astrophysical Journal Letters*, 983, L36 [6 citations]

2024: J. Zhang, D. Huber, L. M. Weiss, J. W. Xuan, J. A. Burt, et al. **(incl. RAR)**, "A Testbed for Tidal Migration: The 3D Architecture of an Eccentric Hot Jupiter HD 118203 b Accompanied by a Possibly Aligned Outer Giant Planet", *The Astronomical Journal*, 168, 295 [15 citations]

2024: J. Lubin, E. A. Petigura, J. Van Zandt, C. Beard, F. Dai, et al. **(incl. RAR)**, "The HD 191939 Exoplanet System is Well Aligned and Flat", *The Astronomical Journal*, 168, 196 [10 citations]

2024: H. Isaacson, A. W. Howard, B. Fulton, E. A. Petigura, L. M. Weiss, et al. **(incl. RAR)**, "The California Legacy Survey. V. Chromospheric Activity Cycles in Main-sequence Stars", *The Astrophysical Journal Supplement Series*, 274, 35 [24 citations]

2024: F. Dai, A. W. Howard, S. Halverson, J. Orell-Miquel, E. Pallé, et al. **(incl. RAR)**, "An Earth-sized Planet on the Verge of Tidal Disruption", *The Astronomical Journal*, 168, 101 [22 citations]

2024: N. Saunders, S. K. Grunblatt, A. Chontos, F. Dai, D. Huber, et al. **(incl. RAR)**, "TESS Giants Transiting Giants. VI. Newly Discovered Hot Jupiters Provide Evidence for Efficient Obliquity Damping after the Main Sequence", *The Astronomical Journal*, 168, 81 [15 citations]

2024: S. W. Yee, E. A. Petigura, H. Isaacson, A. W. Howard, S. Blunt, et al. **(incl. RAR)**, "Additional Doppler Monitoring Corroborates HAT-P-11c as a Planet", *Research Notes of the American Astronomical Society*, 8, 187 [4 citations]

2024: P. A. Dalba, S. R. Kane, H. Isaacson, B. Fulton, A. W. Howard, et al. **(incl. RAR)**, "Giant Outer Transiting Exoplanet Mass (GOT 'EM) Survey. IV. Long-term Doppler Spectroscopy for 11 Stars Thought to Host Cool Giant Exoplanets", *The Astrophysical Journal Supplement Series*, 271, 16 [13 citations]

2024: A. Householder, L. M. Weiss, J. E. Owen, H. Isaacson, A. W. Howard, et al. **(incl. RAR)**, "Investigating the Atmospheric Mass Loss of the Kepler-105 Planets Straddling the Radius Gap", *The Astronomical Journal*, 167, 84 [1 citation]

2023: F. Dai, K. C. Schlaufman, H. Reggiani, L. Bouma, A. W. Howard, et al. **(incl. RAR)**, "A Mini-Neptune Orbiting the Metal-poor K Dwarf BD+29 2654", *The Astronomical Journal*, 166, 49 [16 citations]

2023: E. Knudstrup, D. Gandolfi, G. Nowak, C. M. Persson, E. Furlan, et al. **(incl. RAR)**, "Radial velocity confirmation of a hot super-Neptune discovered by TESS with a warm Saturn-mass companion", *Monthly Notices of the Royal Astronomical Society*, 519, 5637 [13 citations]

2023: F. Dai, K. Masuda, C. Beard, P. Robertson, M. Goldberg, et al. **(incl. RAR)**, "TOI-1136 is a Young, Coplanar, Aligned Planetary System in a Pristine Resonant Chain", *The Astronomical Journal*, 165, 33 [76 citations]

2023: M. El Mufti, P. P. Plavchan, H. Isaacson, B. L. Cale, D. L. Feliz, et al. **(incl. RAR)**, "TOI 560: Two Transiting Planets Orbiting a K Dwarf Validated with iSHELL, PFS, and HIRES RVs", *The Astronomical Journal*, 165, 10 [10 citations]

2022: O. Barragán, D. J. Armstrong, D. Gandolfi, I. Carleo, A. A. Vidotto, et al. **(incl. RAR)**, "The young HD 73583 (TOI-560) planetary system: two 10-$M_\oplus$ mini-Neptunes transiting a 500-Myr-old, bright, and active K dwarf", *Monthly Notices of the Royal Astronomical Society*, 514, 1606 [59 citations]

2022: M. C. Johnson, T. J. David, E. A. Petigura, H. T. Isaacson, J. Van Zandt, et al. **(incl. RAR)**, "An Aligned Orbit for the Young Planet V1298 Tau b", *The Astronomical Journal*, 163, 247 [40 citations]

2022: J. G. Winters, R. Cloutier, A. A. Medina, J. M. Irwin, D. Charbonneau, et al. **(incl. RAR)**, "A Second Planet Transiting LTT 1445A and a Determination of the Masses of Both Worlds", *The Astronomical Journal*, 163, 168 [57 citations]

2022: N. Heidari, I. Boisse, J. Orell-Miquel, G. Hébrard, L. Acuña, et al. **(incl. RAR)**, "HD 207897 b: A dense sub-Neptune transiting a nearby and bright K-type star", *Astronomy and Astrophysics*, 658, A176 [18 citations]

2021: J. Llop-Sayson, J. J. Wang, J.-B. Ruffio, D. Mawet, S. Blunt, et al. **(incl. RAR)**, "Constraining the Orbit and Mass of epsilon Eridani b with Radial Velocities, Hipparcos IAD-Gaia DR2 Astrometry, and Multiepoch Vortex Coronagraphy Upper Limits", *The Astronomical Journal*, 162, 181 [39 citations]

2021: M. Rice, S. Wang, A. W. Howard, H. Isaacson, F. Dai, et al. **(incl. RAR)**, "SOLES I: The Spin-Orbit Alignment of K2-140 b", *The Astronomical Journal*, 162, 182 [33 citations]

2021: J. Zhang, L. M. Weiss, D. Huber, S. Blunt, A. Chontos, et al. **(incl. RAR)**, "Long-period Jovian Tilts the Orbits of Two sub-Neptunes Relative to Stellar Spin Axis in Kepler-129", *The Astronomical Journal*, 162, 89 [26 citations]

2021: L. J. Rosenthal, B. J. Fulton, L. A. Hirsch, H. T. Isaacson, A. W. Howard, et al. **(incl. RAR)**, "The California Legacy Survey. I. A Catalog of 178 Planets from Precision Radial Velocity Monitoring of 719 Nearby Stars over Three Decades", *The Astrophysical Journal Supplement Series*, 255, 8 [359 citations]

2021: B. J. Fulton, L. J. Rosenthal, L. A. Hirsch, H. Isaacson, A. W. Howard, et al. **(incl. RAR)**, "California Legacy Survey. II. Occurrence of Giant Planets beyond the Ice Line", *The Astrophysical Journal Supplement Series*, 255, 14 [283 citations]

2021: M. R. Kosiarek, D. A. Berardo, I. J. M. Crossfield, C. Laguna, C. Piaulet, et al. **(incl. RAR)**, "Physical Parameters of the Multiplanet Systems HD 106315 and GJ 9827", *The Astronomical Journal*, 161, 47 [19 citations]